Linear Algebra and Its Applications

Section 4.8: Applications to Difference Equations

Detailed worked solutions

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Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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4.8 Applications to difference equations

A signal in this section is a sequence $\{y_k\}$ defined for every integer k, including negative integers. For an order-n constant-coefficient homogeneous equation with nonzero first and last coefficients, specifying n consecutive values determines exactly one signal (Theorem 16). Its solution space therefore has dimension n (Theorem 17). To prove that n proposed solutions form a basis, verify the equations when necessary and prove independence. One convenient independence test is the Casorati matrix whose columns are the signals evaluated at n consecutive indices. If that matrix is invertible at one index, the signals are independent. Fewer than n solutions cannot span an n-dimensional space.

Practice problem 4.8.P1: A three-signal basis

The problem already supplies that 2^k, $3^k \sin(k\pi/2)$, and $3^k \cos(k\pi/2)$ solve the third-order equation. Their values at $0, 1, 2$ form the matrix

$$C(0) = \begin{pmatrix} 1 & 0 & 1 \\ 2 & 3 & 0 \\ 4 & 0 & -9 \end{pmatrix}, \qquad \det C(0) = -39 \neq 0.$$

A relation among the three signals would give $C(0)c = 0$, forcing all coefficients to vanish. The solution space has dimension three; these three independent solutions are therefore a basis.

Exercise 4.8.1: Substitute two exponential signals

For $y_k = r^k$, the left side of $y_{k+2} + 2y_{k+1} - 8y_k = 0$ is $r^k(r^2 + 2r - 8) = r^k(r-2)(r+4)$. It vanishes for $r = 2$ and $r = -4$. Both are nonzero, so these signals are defined at negative indices as well. Thus 2^k and $(-4)^k$ are solutions for every integer k.

Exercise 4.8.2: Two roots with opposite signs

For $y_k = r^k$, the equation $y_{k+2} - 25y_k = 0$ becomes $r^k(r^2 - 25) = 0$. The roots $r = 5$ and $r = -5$ make it hold for every integer k, verifying the two proposed solutions.

Exercise 4.8.3: Show the first two verified solutions span

By Exercise 4.8.1 the signals solve the equation. Their values at $k = 0, 1$ form $\left(\begin{smallmatrix} 1 & 1 \\ 2 & -4 \end{smallmatrix}\right)$, with determinant $-6 \neq 0$. They are independent. The equation is of order 2 and has nonzero last coefficient -8, so its solution space has dimension two. Thus the two signals form a basis, and all solutions are $y_k = c_1 2^k + c_2(-4)^k$.

Exercise 4.8.4: A basis for the second equation

The solutions $5^k, (-5)^k$ from Exercise 4.8.2 give initial-value matrix $\left(\begin{smallmatrix} 1 & 1 \\ 5 & -5 \end{smallmatrix}\right)$, whose determinant is -10. Thus they are independent and, since the solution space has dimension two, form a basis. Every solution is $y_k = c_1 5^k + c_2(-5)^k$.

Exercise 4.8.5: A repeated root produces a factor of k

For $u_k = (-2)^k$, the substitution factor is $r^2 + 4r + 4 = (r+2)^2$, so u solves the equation. For $v_k = k(-2)^k$, substitution gives

$$v_{k+2} + 4v_{k+1} + 4v_k = (-2)^k[4(k+2) - 8(k+1) + 4k] = 0.$$

The initial-value matrix $\left(\begin{smallmatrix} 1 & 0 \\ -2 & -2 \end{smallmatrix}\right)$ has determinant $-2 \neq 0$, proving independence. There are two independent solutions in a two-dimensional solution space; hence all solutions are $y_k = (c_1 + c_2 k)(-2)^k$.

Exercise 4.8.6: A cosine-sine pair

For either sine or cosine, adding π to the angle changes its sign. Thus the signals $u_k = 4^k \cos(k\pi/2)$ and $v_k = 4^k \sin(k\pi/2)$ satisfy $u_{k+2} = -16u_k$ and $v_{k+2} = -16v_k$. The initial-value matrix is $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 4 \end{smallmatrix}\right)$ with determinant 4. They are independent and, by the order-two dimension theorem, a basis. Every solution is $4^k[c_1 \cos(k\pi/2) + c_2 \sin(k\pi/2)]$.

Exercise 4.8.7: Test the three supplied solutions by initial values

The exercise permits us to assume the three listed signals solve its order-three equation. Evaluate them at $k = 0, 1, 2$ in the listed order:

$$C(0) = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & -2 \\ 1 & 4 & 4 \end{pmatrix}, \qquad \det C(0) = 12 \neq 0.$$

Therefore the three signals are independent: a relation holding at every index must in particular vanish at these three indices, where the invertible matrix forces all coefficients to be zero. Since the solution space has dimension three, the proposed signals form a basis.

Exercise 4.8.8: Test the three supplied solutions by initial values

The exercise permits us to assume the three listed signals solve its order-three equation. Evaluate them at $k = 0, 1, 2$ in the listed order:

$$C(0) = \begin{pmatrix} 1 & 1 & 1 \\ -1 & 2 & 3 \\ 1 & 4 & 9 \end{pmatrix}, \qquad \det C(0) = 12 \neq 0.$$

Therefore the three signals are independent: a relation holding at every index must in particular vanish at these three indices, where the invertible matrix forces all coefficients to be zero. Since the solution space has dimension three, the proposed signals form a basis.

Exercise 4.8.9: Test the three supplied solutions by initial values

The exercise permits us to assume the three listed signals solve its order-three equation. Evaluate them at $k = 0, 1, 2$ in the listed order:

$$C(0) = \begin{pmatrix} 1 & 1 & 0 \\ 2 & 0 & 5 \\ 4 & -25 & 0 \end{pmatrix}, \qquad \det C(0) = 145 \neq 0.$$

Therefore the three signals are independent: a relation holding at every index must in particular vanish at these three indices, where the invertible matrix forces all coefficients to be zero. Since the solution space has dimension three, the proposed signals form a basis.

Exercise 4.8.10: Test the three supplied solutions by initial values

The exercise permits us to assume the three listed signals solve its order-three equation. Evaluate them at $k = 0, 1, 2$ in the listed order:

$$C(0) = \begin{pmatrix} 1 & 0 & 1 \\ -2 & -2 & 3 \\ 4 & 8 & 9 \end{pmatrix}, \qquad \det C(0) = -50 \neq 0.$$

Therefore the three signals are independent: a relation holding at every index must in particular vanish at these three indices, where the invertible matrix forces all coefficients to be zero. Since the solution space has dimension three, the proposed signals form a basis.

Exercise 4.8.11: Too few solutions for a third-order equation

The equation has order 3 and nonzero last coefficient 4, so its solution space has dimension 3. Only two signals are listed. Even if independent, their span has dimension at most 2, so they cannot form a basis for the whole solution space. The answer is no.

Exercise 4.8.12: Too few solutions for a fourth-order equation

Here the highest and lowest shifts differ by four and the last coefficient is $36 \neq 0$. The solution space therefore has dimension four. Two listed signals cannot span it. Thus they are not a basis, despite being solutions.

Exercise 4.8.13: Two distinct fractional roots

The auxiliary polynomial factors as $r^2 - r + 2/9 = (r - 1/3)(r - 2/3)$. Hence $(1/3)^k$ and $(2/3)^k$ solve the equation. Their initial-value matrix has determinant $2/3 - 1/3 = 1/3 \neq 0$, so they are independent. The dimension-two theorem proves that they span:

$$y_k = c_1(1/3)^k + c_2(2/3)^k.$$

Exercise 4.8.14: Two positive integer roots

The auxiliary polynomial is $r^2 - 5r + 6 = (r - 2)(r - 3)$. The solutions $2^k, 3^k$ have initial-value determinant $3 - 2 = 1 \neq 0$. They are therefore independent; since the solution space has dimension two, all solutions are $y_k = c_1 2^k + c_2 3^k$.

Exercise 4.8.15: A leading coefficient different from one

The auxiliary polynomial is $6r^2 + r - 2 = (2r - 1)(3r + 2)$, with roots $1/2, -2/3$. Dividing the original equation by 6 gives an equivalent order-two equation with nonzero last coefficient, so its solution

space has dimension two. The initial-value determinant is $-2/3 - 1/2 = -7/6 \neq 0$. Thus a basis is $\{(1/2)^k, (-2/3)^k\}$ and every solution is their linear combination.

Exercise 4.8.16: A familiar difference equation

The polynomial $r^2 - 25$ has roots $5, -5$. Their exponential signals solve the equation and have initial-value determinant $-10 \neq 0$. Consequently they form a basis by the dimension-two theorem: $y_k = c_1 5^k + c_2(-5)^k$.

Exercise 4.8.17: Income approaches its equilibrium

For $a = 9/10, b = 4/9$ the equation becomes

$$Y_{k+2} - \tfrac{13}{10} Y_{k+1} + \tfrac{2}{5} Y_k = 1.$$

A constant solution $Y_k = T$ satisfies $(1 - 13/10 + 2/5)T = 1$, giving $T = 10$. The homogeneous polynomial factors as $(r - 4/5)(r - 1/2)$. Its two distinct roots give independent exponential solutions, hence span the homogeneous solution space. Subtracting the particular solution 10 from any nonhomogeneous solution gives a homogeneous solution. Therefore

$$Y_k = 10 + c_1(4/5)^k + c_2(1/2)^k.$$

Both geometric terms tend to zero as $k \to +\infty$, so $Y_k \to 10$, regardless of the fixed initial values.

Exercise 4.8.18: A second stable income model

For $a = 9/10, b = 1/2$, the equation is $Y_{k+2} - (27/20)Y_{k+1} + (9/20)Y_k = 1$. A constant particular solution is again 10, since $1 - 27/20 + 9/20 = 1/10$. The characteristic roots are $3/4$ and $3/5$, as $(r - 3/4)(r - 3/5) = r^2 - (27/20)r + 9/20$. Thus the full solution is

$$Y_k = 10 + c_1(3/4)^k + c_2(3/5)^k.$$

The distinct-root initial-value matrix is invertible, so these terms give every homogeneous solution, not just examples. In particular this model also tends to 10 as k increases.

Exercise 4.8.19: Two real roots for the bending-moment recurrence

For $y_{k+2} + 4y_{k+1} + y_k = 0$, the roots of $r^2 + 4r + 1$ are $r = -2 + \sqrt{3}$ and $s = -2 - \sqrt{3}$. They are distinct and nonzero. The initial-value matrix for r^k, s^k has determinant $s - r = -2\sqrt{3} \neq 0$. The order-two theorem therefore gives the general solution

$$y_k = c_1 r^k + c_2 s^k.$$

These two terms can equivalently be written with exponent $k - 1$ by absorbing one nonzero factor into each arbitrary coefficient.

Exercise 4.8.20: Impose the two boundary values

Assume $N \geq 2$; for $N = 1$ the two stated boundary values would conflict. Write $y_k = Ar^{k-1} + Bs^{k-1}$ using the roots in Exercise 4.8.19. Then $y_1 = 5000$ and $y_N = 0$ give $A + B = 5000$ and $Ar^{N-1} + Bs^{N-1} = 0$. Eliminating one coefficient gives

$$y_k = 5000\,\frac{r^{k-1}s^{N-1} - s^{k-1}r^{N-1}}{s^{N-1} - r^{N-1}}.$$

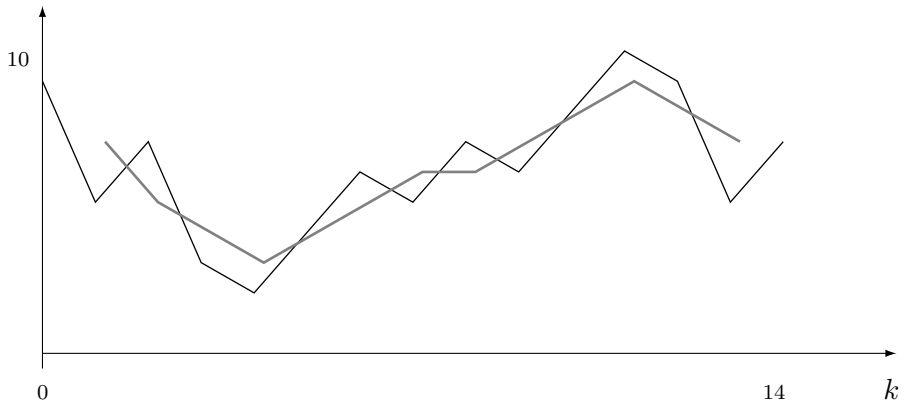
The denominator is nonzero because $|r| \neq |s|$ and $N - 1 > 0$. At $k = 1$ the fraction is 1; at $k = N$ its numerator is zero. Every term solves the recurrence, so their combination does too. Thus this is the required, uniquely determined bending-moment sequence on the support indices.

Exercise 4.8.21: Compute and plot a moving average

Each output is the average of the entry and its two neighbors: $z_k = (y_{k-1} + y_k + y_{k+1})/3$. For instance $z_1 = (9 + 5 + 7)/3 = 7$ and $z_2 = (5 + 7 + 3)/3 = 5$. Continuing through the supplied fifteen entries gives

$$(z_1, \ldots, z_{13}) = (7, 5, 4, 3, 4, 5, 6, 6, 7, 8, 9, 8, 7).$$

The original signal has indices 0 through 14, but the smoothed signal here only has indices 1 through 13; there is no invented endpoint average. In the plot below, the solid broken line is the original signal and the thicker gray broken line is the moving average, using the same axes.



Exercise 4.8.22: A filter separates two frequencies

The filter of Example 3 is $z_k = (\sqrt{2}/4)y_{k+2} + (1/2)y_{k+1} + (\sqrt{2}/4)y_k$. The decimals .35 and .7 in the example abbreviate exact radicals; treating them as rounded decimal inputs would introduce artificial error.

(a) Compute the exact output.

The input is $y_k = 2\cos(k\pi/4) + \cos(3k\pi/4)$. For $u_k = \cos(k\theta)$, angle addition gives

$$u_{k+2} + u_k = 2\cos\theta\cos((k+1)\theta).$$

Thus its filtered output is $[1/2+(\sqrt{2}/2)\cos\theta]\cos((k+1)\theta)$. For $\theta=\pi/4$ the bracket is 1; for $\theta=3\pi/4$ it is 0. Therefore

$$z_k = 2\cos((k+1)\pi/4),$$

with successive values, starting at $k=0$, $\sqrt{2}, 0, -\sqrt{2}, -2, -\sqrt{2}, 0, \sqrt{2}, 2, \ldots$.

(b) Explain the relation to the example.

The filter is linear, so it processes the two input components separately. Example 3 passes the lower-frequency component with a one-index advance and removes the higher-frequency component. The coefficient 2 simply doubles the former output; the latter still contributes zero.

Exercise 4.8.23: A loan paid off by a final smaller payment

(a) Model interest before each payment.

The balance obeys $y_{k+1} = 1.01y_k - 450$, with $y_0 = 10000$, until the last payment. Solving gives $y_k = 45000 - 35000(1.01)^k$ while full payments are made.

(b) A complete balance table and a reproducible program.

The table uses full-precision internal balances and displays dollars rounded to cents; the recurrence itself does not round the balance after each month.

Month	Balance	Month	Balance	Month	Balance
0	10,000.00	9	6,721.02	18	3,134.84
1	9,650.00	10	6,338.23	19	2,716.19
2	9,296.50	11	5,951.61	20	2,293.35
3	8,939.46	12	5,561.12	21	1,866.28
4	8,578.86	13	5,166.74	22	1,434.94
5	8,214.65	14	4,768.40	23	999.29
6	7,846.79	15	4,366.09	24	559.29
7	7,475.26	16	3,959.75	25	114.88
8	7,100.02	17	3,549.34	26	0.00

The following Python code generates the table and adjusts the last payment:

```
from decimal import Decimal, getcontext
getcontext().prec = 50
y = Decimal('10000'); payment = Decimal('450'); k = 0
print(k, f'{y:.2f}')
while y * Decimal('1.01') > payment:
    y = y * Decimal('1.01') - payment
    k += 1
    print(k, f'{y:.2f}')
last = y * Decimal('1.01')
print(k + 1, '0.00', 'last payment', f'{last:.2f}')
```

(c) Find the last payment and the total amount paid.

After payment 25, the unrounded balance is about 114.8801743968. One more month's interest makes the amount due 116.0289761408. Thus the final payment is at month 26, for approximately \$116.03; there have been 25 full \$450 payments. The total paid is $25(450) + 116.0289761408 \approx \$11{,}366.03$. Continuing to subtract a full \$450 after payoff would produce a negative balance and would not model the actual final payment.

Exercise 4.8.24: Monthly deposits and compound interest

(a) Apply interest, then make the new deposit.

The recurrence is $y_{k+1} = 1.005y_k + 200$ with $y_0 = 1000$. Its closed form is $y_k = 41000(1.005)^k - 40000$: substituting it checks both the initial value and the recurrence.

(b) The complete table for months zero through sixty.

Month	Balance	Month	Balance	Month	Balance
0	1,000.00	21	5,527.22	42	10,554.34
1	1,205.00	22	5,754.86	43	10,807.11
2	1,411.02	23	5,983.63	44	11,061.15
3	1,618.08	24	6,213.55	45	11,316.45
4	1,826.17	25	6,444.62	46	11,573.04
5	2,035.30	26	6,676.84	47	11,830.90
6	2,245.48	27	6,910.23	48	12,090.06
7	2,456.71	28	7,144.78	49	12,350.51
8	2,668.99	29	7,380.50	50	12,612.26
9	2,882.33	30	7,617.40	51	12,875.32
10	3,096.75	31	7,855.49	52	13,139.70
11	3,312.23	32	8,094.77	53	13,405.39
12	3,528.79	33	8,335.24	54	13,672.42
13	3,746.43	34	8,576.92	55	13,940.78
14	3,965.17	35	8,819.80	56	14,210.49
15	4,184.99	36	9,063.90	57	14,481.54
16	4,405.92	37	9,309.22	58	14,753.95
17	4,627.95	38	9,555.77	59	15,027.72
18	4,851.09	39	9,803.55	60	15,302.86
19	5,075.34	40	10,052.56		
20	5,300.72	41	10,302.83		

Here is the program used, retaining internal precision until display:

```
from decimal import Decimal, getcontext
getcontext().prec = 50
y = Decimal('1000')
for k in range(61):
    print(k, f'{y:.2f}')
    y = y * Decimal('1.005') + Decimal('200')
```

(c) Separate deposits from interest.

At months $24, 48, 60$, the balances are respectively \$6,213.55, \$12,090.06, and \$15,302.86, rounded to cents. By month 60 the deposits total $1000 + 60(200) = 13000$. Thus interest accounts for $15302.8562545 - 13000 \approx \$2{,}302.86$.

Exercise 4.8.25: A quadratic particular solution

For $y_k = k^2$, substitution gives

$$(k+2)^2 + 3(k+1)^2 - 4k^2 = 10k + 7,$$

which is the required right-hand side. The homogeneous polynomial is $r^2 + 3r - 4 = (r-1)(r+4)$, with independent exponential solutions $1, (-4)^k$. Any two particular solutions differ by a homogeneous solution, so the general solution is $y_k = k^2 + c_1 + c_2(-4)^k$.

Exercise 4.8.26: A linear particular solution

For $y_k = 1 + k$, the left side is $(k+3) - 6(k+2) + 5(k+1) = -4$. The homogeneous roots of $r^2 - 6r + 5$ are $1, 5$, so its solutions are $c_1 + c_2 5^k$. Hence all solutions are $y_k = 1 + k + c_1 + c_2 5^k$. The constant 1 could be absorbed into c_1, but retaining it makes the supplied particular solution visible.

Exercise 4.8.27: Another linear particular solution

For $y_k = k - 2$, compute $y_{k+2} - 4y_k = k - 4(k-2) = 8 - 3k$. The homogeneous roots are $2, -2$. Thus the full solution is $y_k = k - 2 + c_1 2^k + c_2(-2)^k$. The two exponential solutions are independent because their values at indices zero and one have determinant $-4 \neq 0$.

Exercise 4.8.28: An affine particular solution with roots of magnitude five

For $y_k = 1+2k$, the left side is $(1+2(k+2)) - 25(1+2k) = -48k - 20$, exactly as required. The associated homogeneous equation has roots $5, -5$. Therefore the general solution is $y_k = 1 + 2k + c_1 5^k + c_2(-5)^k$.

Exercise 4.8.29: A fourth-order equation as a first-order vector recurrence

Choose the state $x_k = (y_k, y_{k+1}, y_{k+2}, y_{k+3})^T$. The first three entries of x_{k+1} just shift those coordinates left. Solving the recurrence for y_{k+4} gives $y_{k+4} = 2y_k - 6y_{k+1} + 8y_{k+2} - 3y_{k+3}$. Thus

$$x_{k+1} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 2 & -6 & 8 & -3 \end{pmatrix} x_k.$$

The state definition is part of the answer: reversing the order of its coordinates would change the companion matrix.

Exercise 4.8.30: A third-order companion system

Set $x_k = (y_k, y_{k+1}, y_{k+2})^T$. Since $y_{k+3} = 5y_{k+2} - 8y_k$, the desired system is

$$x_{k+1} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -8 & 0 & 5 \end{pmatrix} x_k.$$

The middle coefficient is zero because there is no y_{k+1} term in the original equation.

Exercise 4.8.31: Remove an irrelevant common shift

The equation $y_{k+3} + 5y_{k+2} + 6y_{k+1} = 0$ has no y_k term. Put $j = k + 1$. Because the equation holds for *all* integers k, it is equivalent to $y_{j+2} + 5y_{j+1} + 6y_j = 0$ for all integers j. Its order is therefore 2, not 3: the largest and smallest shifts differ by two and both endpoint coefficients are nonzero.

Exercise 4.8.32: The order depends on the last nonzero coefficient

For $y_{k+3} + a_1y_{k+2} + a_2y_{k+1} + a_3y_k = 0$: if $a_3 \neq 0$ the order is 3; if $a_3 = 0, a_2 \neq 0$ a shift makes the order 2; if $a_3 = a_2 = 0, a_1 \neq 0$ it is 1. If all three are zero, it says $y_{k+3} = 0$ at every integer, which is just the zero-signal condition, of order 0 in this shift-width convention. Without a nonzero hypothesis on the final coefficient, one cannot simply call every equation in the displayed form third order.

Exercise 4.8.33: Independence does not require an invertible Casorati sample for arbitrary signals

Let $y_k = k^2$ and $z_k = 2k|k|$. If $ay_k + bz_k = 0$ for every integer k, using $k = 1$ gives $a + 2b = 0$ and using $k = -1$ gives $a - 2b = 0$. Subtracting gives $b = 0$, then $a = 0$. Thus the signals are independent on all integers (they would be dependent if restricted to nonnegative indices). The requested matrices are

$$C(0) = \begin{pmatrix} 0 & 0 \\ 1 & 2 \end{pmatrix}, \quad C(-1) = \begin{pmatrix} 1 & -2 \\ 0 & 0 \end{pmatrix}, \quad C(-2) = \begin{pmatrix} 4 & -8 \\ 1 & -2 \end{pmatrix}.$$

All are singular. Invertibility of one Casorati matrix is a sufficient independence test, but its failure alone does not prove dependence for arbitrary signals. No common second-order recurrence hypothesis was given here.

Exercise 4.8.34: Sampling can destroy independence

Take $f(t) = 1$, $g(t) = t$, and $h(t) = \sin(\pi t)$. They are independent functions: a relation $a + bt + c\sin(\pi t) = 0$ for every real t, evaluated at $0, 1, 1/2$, successively gives $a = b = c = 0$. But sampling at integers turns h into the zero signal. The three sampled signals are therefore dependent. Thus the answer is no: distinct continuous behaviors can become indistinguishable at the chosen sample points.

Exercise 4.8.35: A linear difference operator on signals

For two signals u, v and a scalar c, evaluate each index:

$$\begin{aligned}(T(u+v))_k &= (u_{k+2}+v_{k+2}) + a(u_{k+1}+v_{k+1}) + b(u_k+v_k)\\ &= (Tu)_k + (Tv)_k,\\ (T(cu))_k &= cu_{k+2} + acu_{k+1} + bcu_k = c(Tu)_k.\end{aligned}$$

The resulting sequences are defined at every integer, so T maps S into S. The two coordinate-wise identities prove linearity as an equality of signals.

Exercise 4.8.36: Add a homogeneous solution to a particular solution

Suppose $T(x_p) = z$ and $u \in \ker T$, so $T(u) = 0$. Linearity gives $T(u+x_p) = T(u)+T(x_p) = 0+z = z$. Thus $u + x_p$ solves the same nonhomogeneous equation. Conversely, if $T(x) = z$, then $T(x - x_p) = z - z = 0$, explaining why all solutions are precisely a particular solution plus kernel vectors.

Exercise 4.8.37: One-sided shifts have only a one-sided inverse

Apply the maps directly to an arbitrary sequence $y = (y_0, y_1, \ldots)$:

$$TD(y) = T(0, y_0, y_1, \ldots) = (y_0, y_1, \ldots) = y.$$

Hence $TD = I$. But $DT(y) = D(y_1, y_2, \ldots) = (0, y_1, y_2, \ldots)$, which need not equal y. For example $y = (1, 0, 0, \ldots)$ is sent to the zero sequence by DT. Thus $DT \neq I$, illustrating why the finite-dimensional square-matrix conclusion about one-sided inverses does not extend to this infinite-dimensional space.